

The Highway Analogy

Before diving into definitions, let us use a simple analogy. Think of a highway connecting two cities:

- **Bandwidth** is the number of lanes on the highway. More lanes mean more cars can travel simultaneously.
- **Throughput** is how many cars actually pass through per hour. This depends on traffic conditions, not just the number of lanes.
- **Latency** is the time it takes for a single car to travel from one city to the other.

A highway might have 4 lanes (high bandwidth), but if there is an accident, only 100 cars per hour pass through (low throughput). Meanwhile, each car might take 2 hours to complete the journey (high latency).

Latency

Latency is the time it takes for a single request to travel from source to destination and back. It measures delay. In networking, latency is often called **round-trip time (RTT)**, the time from sending a request to receiving a response.

ServerClientServerClientLatency = Total Round-Trip TimeProcessing(50ms)Total Latency: 150msRequest (t=0ms)Response (t=150ms)

Components of Latency

Latency is not a single value. It is the sum of multiple delays:

Client

Network
Latency

Server
Processing

Database
Query

Response
Assembly

Network
Latency

Client

1. **Propagation delay:** Time for signals to travel through the medium. Light in fiber travels at ~200,000 km/s. A cross-Atlantic request (6,000 km) takes ~30ms just for propagation.
2. **Transmission delay:** Time to push bits onto the wire. Depends on packet size and link bandwidth.
3. **Processing delay:** Time for routers, load balancers, and servers to process packets.
4. **Queuing delay:** Time spent waiting in buffers when components are busy.

Measuring Latency

Latency is typically measured using percentiles:

Metric	Description
p50 (median)	50% of requests are faster than this
p95	95% of requests are faster than this
p99	99% of requests are faster than this
p99.9	99.9% of requests are faster than this

Why percentiles matter: Average latency hides outliers. A system with 10ms average might have p99 of 500ms, meaning 1% of users experience terrible performance.

What Affects Latency?

Factor	Impact
Geographic distance	More distance = more propagation delay
Network congestion	Causes queuing delays
Server load	Increases processing time
Database queries	Slow queries add latency
DNS resolution	Cold requests need DNS lookup
TLS handshake	Adds 1-2 round trips

Reducing Latency

- **Use CDNs:** Serve content from edge locations closer to users
- **Caching:** Eliminate round trips by caching at multiple layers
- **Connection pooling:** Avoid repeated connection setup

- **Database optimization:** Add indexes, optimize queries
- **Geographic distribution:** Deploy servers closer to users
- **Protocol optimization:** Use HTTP/2, HTTP/3 (QUIC)

Throughput

Throughput is the amount of work completed per unit of time. It measures volume. For web systems, throughput is often expressed as **requests per second (RPS)** or **transactions per second (TPS)**.

Throughput vs Bandwidth

A common confusion: **bandwidth** is theoretical maximum capacity, while **throughput** is actual achieved rate.

Metric	Definition	Example
Bandwidth	Maximum possible data transfer rate	1 Gbps network link
Throughput	Actual data transfer achieved	600 Mbps actual transfer

You can never have throughput higher than bandwidth, but throughput is almost always lower due to:

- Protocol overhead (headers, acknowledgments)
- Congestion and packet loss
- Processing limitations
- Inefficient resource utilization

Calculating Throughput

For a single-threaded system:

For a multi-threaded system:

What Limits Throughput?

Throughput Limiters

CPU
Compute-bound tasks

Memory
Large working sets

I/O
Disk or Network

Connections
Connection limits

Contention
Locks, shared resources

The **bottleneck** determines maximum throughput. A system is only as fast as its slowest component.

Improving Throughput

- **Horizontal scaling:** Add more servers
- **Vertical scaling:** Add more CPU, memory
- **Async processing:** Do not block on slow operations
- **Batching:** Process multiple items together
- **Caching:** Reduce work by reusing results
- **Connection pooling:** Reuse expensive connections
- **Load balancing:** Distribute work evenly

Bandwidth

Bandwidth is the maximum rate at which data can be transferred. It measures capacity. Bandwidth is typically expressed in **bits per second (bps)**: Kbps, Mbps, Gbps.

Types of Bandwidth

Type	Description
Network bandwidth	Capacity of network links (1 Gbps Ethernet)
Memory bandwidth	Rate of data transfer to/from RAM (DDR4: ~25 GB/s)
Disk bandwidth	Read/write speed of storage (SSD: ~500 MB/s)
Bus bandwidth	Internal data transfer rate (PCIe 4.0: ~64 GB/s)

Bandwidth-Delay Product

An important concept that connects bandwidth and latency:

BDP represents how much data can be "in flight" at any moment.

Example:

- Bandwidth: 1 Gbps = 125 MB/s
- Latency: 100ms (coast-to-coast US)
- BDP: $125 \text{ MB/s} \times 0.1 \text{ s} = 12.5 \text{ MB}$

This means 12.5 MB of data can be traveling through the pipe at any instant. If your TCP window size is smaller than BDP, you will not fully utilize available bandwidth.